

Troy Mayer

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EDUCATION

Purdue University, School of Mechanical Engineering and School of Science

West Lafayette, IN

B.S. Mechanical Engineering; B.S. Applied Statistics; Minor in Economics

May 2028

Clubs: Club Swimming, House Band

GPA:3.4/4

PROFESSIONAL EXPERIENCE

Boston Dynamics

Waltham, MA

Sensors and Perception Co-Op – Atlas

July 2026 – Present

- Built a forward-kinematics pipeline propagating 6-DOF floating-base poses through multi-joint kinematic chains using roll-pitch-yaw matrices and Rodrigues' rotation formula, deriving end-effector velocities by rigid-body velocity composition.
- Designed a 4th-order zero-phase Bessel anti-alias filter for impact analysis that captures 85% of a 3 ms impact peak (vs. 30% at 100 Hz), selected over Butterworth by measurement with under 1.3% error across 3–30 ms pulses.
- Built and deployed a Flask/Plotly forensics dashboard on a Linux VM analyzing ~28k impact events from ~900 test logs, with an ETL pipeline into a SQLite cache, a 3D impact heatmap, and per-event waveform inspection.
- Modeled hardware thermal behavior in Python (Pandas, SciPy), applying Savitzky-Golay smoothing and regime detection within heating cycles to fit exponential segments via multi-start least squares; projected steady-state temperatures across hundreds of units.

Caterpillar (via Purdue Data Mine)

West Lafayette, IN

Machine Learning Researcher

January 2026 – May 2026

- Built an LSTM time-series model forecasting next-day primer, topcoat, and solvent consumption from five years of paint-line data, achieving a 5% reduction in material usage.
- Developed Monte Carlo simulations over thousands of iterations to attach confidence intervals to each forecast and quantify variance under uncertainty.
- Built Python pipelines to ingest, clean, and consolidate five years of noisy legacy data into one daily series, contributing to a 4% reduction in manufacturing cycle time.

Chortos Lab – Soft Adaptive Systems and Structures

West Lafayette, IN

Undergraduate Researcher

August 2025 – May 2026

- Built data acquisition systems in LabVIEW and MATLAB, processing and normalizing datasets of 50,000+ points to quantify sub-millimeter material behavior.
- Wrote a Python computer-vision tracker resolving cantilever displacement to 0.01 mm, guiding gel formulations that increased force output by over 50%.

PROJECTS

Purdue ASME – Biomechanics

West Lafayette, IN

Active Orthosis – Electronics Team Lead

April 2024 – Present

- Wrote safety-critical C++ firmware for an ESP32-S3 running a 1 kHz PI velocity loop from EMG input to motor command, with 10 fault monitors, a hardware watchdog, and a kill switch.
- Cut EMG-to-motion latency from 103 to 43 ms using a Bayesian EMG estimator, validated in closed-loop simulation on real EMG with 26 automated tests in CI.
- Led a 5-engineer team and redesigned the 4-channel EMG board in KiCad with an INA333 front end, right-leg drive, and ISO7763 motor isolation, raising cross-day gesture decoding from 31% to 87% in emulation.

Legged Robot State Estimation

Personal Project

August 2026 – Present

- Built a contact-aided error-state EKF in C++/Eigen fusing IMU and leg kinematics at 1 kHz, achieving 0.013 m/s velocity RMSE over a 60-second walk vs. 1.38 m/s for IMU-only dead reckoning.
- Verified filter Jacobians against finite differences in GoogleTest and characterized observability, showing roll, pitch, and velocity stay bounded while position and yaw drift.

SKILLS & INTERESTS

- **Tools:** C++, Python, MATLAB, SQL, LabVIEW, CAD (NX, Fusion, Creo), FEA Analysis, KiCad, Snowflake, Git, Linux, Eigen, MuJoCo
- **Languages:** English (Native), Turkish (Fluent), Spanish (Fluent - New Jersey Seal of Biliteracy).
- **Hobbies:** Lifting, Hiking, Guitar, Saxophone